# Regenerative agriculture by the numbers: a visualization-ready data brief

**Healthy soil stores 3× more carbon than the atmosphere, holds thousands of additional gallons of water per percentage point of organic matter gained, and can make farms 78% more profitable — yet industrial agriculture has stripped 50–70% of cultivated soils' original carbon and driven aquifer depletion to crisis levels.** This data brief compiles the most credible, specific, and recent statistics across three pillars — soil carbon, farm economics, and water security — curated for The Carbon Underground and The Edison Institute to translate into interactive web visualizations. Every figure below is sourced from peer-reviewed journals, government agencies, or established research institutions, and prioritized for temporal trends, comparison potential, and geographic specificity.

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## PILLAR 1: THE CARBON BENEATH OUR FEET

### Global soil holds more carbon than the atmosphere and all vegetation combined

The world's soils contain roughly **2,500 gigatons of carbon** (Gt C) in the top two meters — approximately **3.1× the carbon in the atmosphere** (~830 Gt C) and **4× the carbon in all living plants and animals** (~540–650 Gt C). Nature The ocean dwarfs both at ~38,400 Gt C, but soil is the largest *terrestrial* carbon pool by far. Nature About 75% of all terrestrial carbon is stored belowground (Hawkins et al., 2023, *Current Biology*). Global cropland alone stores over **140 Pg C in the top 30 cm**, representing roughly 10% of the total global soil organic carbon (SOC) pool Nature across just 15.9 million km² (Zomer et al., 2017, *Scientific Reports*).

**Visualization-ready carbon pool comparison:**

| Carbon Pool | Stock (Gt C) | Source |
|---|---|---|
| Soils (0–2 m) | ~2,500 | FAO; Lal (2008) |
| Atmosphere | ~830 (2025) | IPCC; updated |
| Vegetation | ~540–650 | FAO |
| Oceans | ~38,400 | Houghton (2007) |
| Pre-industrial atmosphere | ~560–578 | IPCC |

Annual fluxes offer powerful animation potential: plant photosynthesis absorbs **~110 Gt C/year**, soil respiration releases **~61–62 Gt C/year**, K-state and fossil fuel emissions add **~10 Gt C/year**. Soils currently remove roughly **25% of fossil fuel emissions** each year (IPCC data, cited by Columbia Climate School). Columbia Climate School Mycorrhizal fungi alone channel **13.12 Gt $CO_2$-equivalent per year** from plants into underground mycelium networks — equivalent to **36% of annual fossil fuel emissions** ScienceDirect Cell Press (Hawkins et al., 2023, *Current Biology*, based on ~200 datasets from 61 papers). Roughly 90% of all land plants form partnerships with these fungi. Eos

### Industrial agriculture created a massive soil carbon debt

The landmark Sanderman, Hengl & Fiske study (2017, *PNAS*, Woods Hole Research Center) quantified a **133 Pg C "soil carbon debt"** — carbon lost from the top 2 meters of soil over 12,000 years of human land use, PNAS equivalent to ~488 Gt $CO_2$. The rate of loss accelerated dramatically in the past 200 years. Google Sites Intensively managed agricultural soils have lost **50–70% of their pre-cultivation carbon** CropWatch (Lal, 2007; Zomer et al., 2017). A meta-analysis found a **median SOC loss of 26%** in the upper 30 cm when native land converts to agriculture, and **16% loss across the top 100 cm** PNAS (Sanderman et al., 2017). Forest-to-agriculture conversion caused average SOC losses of **52% in temperate regions**, 41% in tropical and 31% in boreal zones Nature (Wei et al., 2014, *Scientific Reports*).

Climate change compounds the problem. A 2023 study in *Nature Scientific Reports* found that climate change alone (1919–2018) caused an additional **2.5 ± 2.3 Mg C/ha loss** (~3.9%) in global agricultural topsoils — independent of management practices. Nature

This historical loss also represents opportunity. The sink capacity of degraded agricultural soils is **50–66% of historic losses** Science (Lal, 2004, *Science*), meaning soils could theoretically reabsorb 42–78 Gt C with proper management. Nature

### Regenerative practices rebuild soil carbon at measurable rates

The global potential for soil carbon sequestration through improved cropland management ranges from **0.28–** PubMed Central **1.85 Pg C/year** depending on practice adoption scenarios (Lessmann et al., 2022, *Global Change Biology*; Zomer et al., 2017, *Scientific Reports*). The widely discussed "4 per 1000" initiative targets a 0.4% annual increase in global soil carbon stocks, Nature equating to roughly **0.6 t C/ha/year** (Minasny et al., 2017, *Geoderma*). Realistic measured rates sit at **0.2–0.5 t C/ha/year** globally. ScienceDirect

**Practice-specific sequestration data:**

- **Cover cropping** sequesters **0.32–** ScienceDirect **0.88 Mg C/ha/year** in the top 30 cm, increasing SOC by **7.3–** Wiley Online Library **15.5%** ScienceDirect (Poeplau & Don, 2015; Joshi et al., 2023; Abdalla et al., 2019). If all 36.4 million hectares of US corn adopted cover crops, an estimated **107 million metric tons $CO_2$-equivalent** could be sequestered annually Wiley Online Library (Joshi et al., 2023, *Agronomy Journal*).
- **Compost application** delivers striking results. The Marin Carbon Project found that a **single application of ½ inch of compost** on rangeland increased forage production by **40–70%** and sequestered **~1 ton C/ha/year**, Naturebasedclimate projected to continue for **30–100 years** Marincarbonproject (Ryals & Silver, 2013, *Ecological Applications*, UC Berkeley). The Carbon Cycle Institute has completed 137 Carbon Farm Plans covering 77,440 acres, projecting **1,793,029 metric tons $CO_2$-equivalent** sequestered over 20 years. Gogreen-project
- **Adaptive multi-paddock (AMP) grazing** stores **13% more soil carbon** ScienceDirect (0–1 m depth) than conventional grazing, with **25% more carbon in stable mineral-associated forms** Rootssodeep (Mosier et al., 2021, *Journal of Environmental Management*). White Oak Pastures in Georgia sequestered **2.29 Mg C/ha/year** over 20 years Regeneration International — enough to fully offset beef production emissions (Rowntree et al., 2020).
- **No-till** shows nuanced results. It increases surface (0–10 cm) carbon but evidence for whole-profile gains is mixed. Net sequestration approaches positive territory after **~14 years** of continuous practice ScienceDirect (Meurer et al., 2022, *Geoderma*). The practice is best viewed as reducing erosion with carbon as a co-benefit. Nature A critical finding from Michigan State University's 30-year study: **a single tillage pass** on never-farmed soil reduces aggregation to levels equivalent to 50+ years of continuous tillage. MSU Extension

**The Rodale Institute Farming Systems Trial** (est. 1981, the longest-running side-by-side comparison) Farm Progress provides benchmark data: after 34 years, the organic-manure system held **61.9 Mg C/ha** versus **53.3 Mg C/ha** in the conventional system (16% higher) in the 0–30 cm layer Wiley Online Library (Lorenz, 2025, *Soil Science Society of America Journal*). Organic systems used **45% less energy** and released **40% fewer carbon emissions**, Beyond Pesticides while producing **31% higher yields during drought years**. CFSPH

### Carbon distribution by depth reveals hidden storage potential

Soil carbon profiles offer rich vertical-visualization opportunities. In the landmark Jobbágy & Jackson (2000) analysis of 2,700+ profiles, the percentage of total carbon in the top 20 cm varied dramatically by ecosystem: **50% in forests, 42% in grasslands, and 33% in shrublands**. ADS Adding the 1–2 m layer increases the global SOC budget by **33%**; the 2–3 m layer adds another **23%**. Unh A 2025 meta-analysis (Wang et al., *Earth System Science Data*) found that SOC density in croplands drops **six-fold** from 30 Mg C/ha (0–20 cm) to 5 Mg C/ha (80–100 cm). Copernicus Global subsoil SOC stock (20–100 cm) totals **803 Pg C**, with forestland holding 547, grassland 181, and cropland 74 Pg C. ADS

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## PILLAR 2: THE ECONOMICS OF SOIL NEGLECT AND RENEWAL

### Soil degradation costs the global economy trillions annually

The most widely cited figure pegs global soil erosion costs at **~$400 billion/year** from the loss of 75 billion tonnes of soil annually (ELD Initiative, 2015; cited by FAO, UNCCD). Land Portal But broader estimates of ecosystem service losses from land degradation reach **$6.3–$10.6 trillion/year**, equivalent to **10–17% of global GDP** (ELD Initiative, 2015). Land Portal The UNCCD warns of **$23 trillion in cumulative losses by 2050** without action. SDG Knowledge Hub Up to **40% of Earth's land is already degraded**, threatening roughly half of global GDP (~$44 trillion) (UNCCD Global Land Outlook 2, 2022). UNCCD

In the United States, a revealing 2020 study by Jang et al. (*Earth's Future*) found that US corn farmers spend **$500+ million/year in extra fertilizer** simply to compensate for soil fertility losses — one-third of all fertilizer applied to corn replaces what degradation took away. ScienceDaily When including nutrient runoff costs and ecosystem impacts like the Gulf of Mexico Dead Zone, total US soil degradation costs exceed **$1 trillion/year** (Neff et al., CU Boulder, 2021). ScienceDaily The economics of inaction are clear: every **$1 invested in reversing land degradation returns $5** over a 30-year horizon (Nkonya et al., 2016). Springer

### Regenerative farms earn more despite sometimes producing less

The landmark LaCanne & Lundgren study (2018, *PeerJ*) compared 20 farms across the Northern Plains and found regenerative operations earned **78% higher profits** despite **29% lower grain production**. Lower input costs and diversified income streams drove the gap, and soil organic matter proved a stronger predictor of profitability than yield. Acresusa Subsequent research has reinforced this finding: regenerative almond orchards in California were **twice as profitable** as conventional counterparts (Fenster, Oikawa & Lundgren, 2021). BCG and WBCSD estimated **15–25% profitability increases** from regenerative transitions (2023). California State University, Chico The American Farmland Trust documented an average **176% ROI** across four study farms (2019). Ecosystems United McKinsey's 2024 analysis projected that farms using no-till plus cover cropping could achieve **10–30% higher yields** and **$20–$60/acre in additional annual income**, with potential for **$250 billion in incremental economic value** over a decade at 80% US corn/soy adoption. McKinsey & Company

Input cost savings of **25–50%** over time from reduced synthetic fertilizer and pesticide dependence are consistently documented. Keystone Bio Ag A California mentor farmer (Daniel Unruh, Colusa County) achieved a **$425.19/acre net income increase** over six years through regenerative cover cropping (CRARS/CSU Chico). California State University, Chico During drought, the advantage widens sharply: McKinsey found regenerative farms maintain **>95% of usual yields** while conventional farms drop to **one-third** of typical production. McKinsey & Company

### Climate change is already cutting crop yields and will accelerate losses

Observed impacts are no longer projections. From 1980–2008, climate change decreased global **corn production by 3.8%** and **wheat by 5.5%** Foodsecurityleadership (Lobell et al., 2011). PubMed Central Over 50 years, net effects have been **negative for barley (−6.4%), maize (−4.2%), and wheat (−3.6%)** (PNAS, 2025). PNAS A recent *Nature* study (Hultgren et al., 2025) quantified the cost at **554 trillion kilocalories lost per 1°C of warming** — roughly 120 kcal per person per day per degree. Carbon Brief

Projections intensify the picture. NASA/AgMIP model ensembles project **maize yields declining 24% by late century** under current trends (Jägermeyr et al., 2021, *Nature Food*). NASA Scientific Visualization … Under worst-case warming (RCP8.5), global yields decline **~24%**; even under Paris Agreement targets, they drop **~6%**. Our World in Data The IPCC projects cereal prices rising by **up to 129%** by 2050 under moderate scenarios, with **1–183 million additional people** at risk of hunger IPCC (IPCC Special Report on Climate Change and Land). Climate-driven food price inflation could add **0.92–3.23 percentage points per year** globally, Nature with **1.4–1.8 points** in North America by 2035 Climate Central (Kotz et al., 2023, *Communications Earth & Environment*).

### Farm financial stress is rising sharply

US Chapter 12 farm bankruptcy filings tell a stark story of reversal: after falling to a historic low of **139 filings in 2023**, Ag Proud they surged **55% to 216 in 2024** American Farm Bureau Federat… and another **46% to 315 in 2025** (US Courts; American Farm Bureau Federation). American Farm Bureau Federat… Total farm debt is projected to reach a **record $624.7 billion in 2026** (USDA). American Farm Bureau Federat… Farm interest expenses rose **43% from 2022 to 2023** Ag Proud and remain **73% above 2018 levels**. American Farm Bureau Federat… Net farm income fell **nearly 40% from 2022 to 2024**. American Farm Bureau Federat… Corn prices have crashed **~50%** since 2022; soybeans are down **~40%**. Fortune Over **140,000 farms disappeared** between the 2017 and 2022 Census of Agriculture. American Farm Bureau Federat…

Extreme weather compounds the pressure. In 2023, US crop and rangeland losses exceeded **$21.9 billion** (AFBF), representing 23.6% of NOAA's $92.9 billion total disaster costs. American Farm Bureau Federat… FarmWeekNow The FAO estimates **$3.8 trillion in global crop and livestock losses** from disasters over the last 30 years FAO (~$123 billion/year, or 5% of global agricultural GDP), Solutionsfromtheland with drought causing over 65% of those losses. FAO The frequency of billion-dollar disasters has accelerated from an average gap of **82 days in the 1980s** to just **18 days in 2018–2022** (NOAA). NOAA Climate.gov

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## PILLAR 3: WHEN THE WATER RUNS OUT

### Healthy soil acts as a water reservoir — but the numbers need precision

The USDA-NRCS popularized the claim that every 1% increase in soil organic matter helps soil hold **20,000 additional gallons of water per acre**. The math checks out at face value (Dr. Michelle Wander, University of Illinois, verified the calculation), but this figure represents *total* water held, not plant-available water. Peer-reviewed analysis using Minasny & McBratney (2017, *Geoderma*) data finds the plant-available water increase is closer to **3,400–10,000 gallons per acre per 1% SOM increase**, depending on soil texture. For visualization purposes, the defensible range is **3,400–20,000 gallons per acre** with clear labeling of what each number represents. Hudson's 1994 landmark study showed that silt loam with 4% SOM holds **more than twice the water** of silt loam with 1% SOM.

Regenerative practices dramatically improve water infiltration. A meta-analysis of 89 field studies (Basche & DeLonge, 2019, *PLOS ONE*) found cover crops increased infiltration rates by **~59%**, no-till by **~45%**, and perennial systems by **~60%** compared to conventional controls. PubMed Central PLOS Cover crops reduced erosion by **>94%** compared to bare soil ScienceDaily (Laloy & Bielders, 2010) and runoff by **up to 90%** in some studies. Farmdoc Daily Ground cover management across 85 studies reduced soil loss by **70.5%** and nitrogen/phosphorus runoff losses by **53–57%**. ScienceDirect

### The Ogallala and Central Valley aquifers approach critical thresholds

The Ogallala (High Plains) Aquifer — spanning 175,000 square miles across eight states USGS and supporting a **$20+ billion agricultural industry** — has lost **286.4 million acre-feet** of recoverable water since predevelopment (~1950), with an area-weighted average decline of **16.5 feet** USGS (USGS, 2024). Wells extract water at **3–50× the recharge rate**, and only about 1 inch of precipitation reaches the aquifer annually. The University of Texas projects **up to 70% of the Texas Panhandle portion** will become unusable within 20 years. Once depleted, natural replenishment would take **over 6,000 years**.

**Kansas data offers compelling time-series:** Southwest Kansas aquifer levels fell **1.52 feet** in just the year between January 2024 and January 2025, exceeding the prior year's 1.43-foot decline (Kansas Geological Survey). Farmers need to cut pumping by **17.5%** to stabilize some areas.

California's Central Valley aquifer faces equally dire metrics. Land subsidence averages **nearly 1 inch per year** across the San Joaquin Valley (2006–2022), with some areas sinking **more than 1 foot annually** (Stanford University, 2024). In the Tule subbasin, ground has sunk **more than 7 feet since 2015**. Historic total subsidence reaches **up to 28 feet** in parts of the valley since the 1920s (USGS). This subsidence has reduced Friant-Kern Canal flow by **60%** — a canal serving **1 million acres and 250,000+ people**. The California Department of Water Resources warned in May 2025 that without action, State Water Project deliveries could decline by **up to 87%** by 2043, dropping from 2.2 million acre-feet to just **295,000 acre-feet**. Permanent aquifer storage loss from compaction equals roughly **the volume of Lake Mead** (NASA, 2024).

### California's Central Valley faces an agricultural reckoning

The Central Valley produces **over 50% of the nation's fruits, nuts, and vegetables** within a **$50 billion agricultural industry** employing more than **400,000 people** Farmwater (California Farm Water Coalition). The Public Policy Institute of California (November 2024) estimates up to **900,000 acres** could be forced out of production, reducing farm revenue by **up to $2.5 billion annually** Farmwater and eliminating **50,000 jobs**. Farmwater UC Davis/UC Merced research (2025) warns that without coordinated action, water supply reductions could fallow **3 million acres**, destroy **67,000 jobs**, UC Davis and cause **$4.3 billion** in direct losses from a 5-million-acre-foot shortage. Waterblueprintca

Almond acreage illustrates the tension: plantings surged from **640,000 to 1,640,000 acres** between 2004 and 2021, consuming **4.7–5.5 million acre-feet/year** — enough water to supply 15–17 million households C-win (C-WIN, 2024). Each almond requires roughly **12 liters of water**. Valuingwaterinitiative SGMA implementation will require additional fallowing of **up to 500,000 acres by the early 2040s**, C-win yet in Kings County, groundwater extraction has **not decreased** since 2015 despite the law. CalMatters

### Global water scarcity will reshape food production

Agriculture consumes **70% of global freshwater withdrawals** UN-Water (FAO) and **92% of humanity's total water footprint** when including rain-fed water (Hoekstra, 2012, *PNAS*). In the US, agriculture accounts for **80% of all water consumed**. FoodPrint By 2050, **5 billion people** — roughly two-thirds of the world's population — will face at least one month of water shortage (WMO). Scientific American Currently, **4 billion people** already experience severe water scarcity at least one month per year UN-Water (Mekonnen & Hoekstra, 2016, *Science Advances*).

Global groundwater depletion has more than doubled, from **126 km³/year in 1960 to 283 km³/year in 2000** (Wada et al., 2010, *Geophysical Research Letters*). A 2024 study in *Nature* (Jasechko & Perrone) found groundwater is dropping in **71% of nearly 1,700 aquifers** studied globally, with depletion rates **accelerating** since 2000 in most cases. Ninety percent of aquifers with accelerating declines are in places that have gotten drier over 40 years. Meanwhile, the world must produce **56% more food calories by 2050** Nature World Resources Institute (WRI), and **60% of irrigated agriculture** already faces extremely high water stress. World Resources Institute Failing to manage water better could cause GDP losses of **7–12% in India, China, and Central Asia** by 2050. World Resources Institute

As of March 3, 2026, **49.5% of the US** is in drought (Drought.gov). Billion-dollar weather disasters now average **23 per year** (2020–2024) versus **9 per year** historically (NOAA). NOAA NOAA Climate.gov Cumulative US disaster costs since 1980 exceed **$2.66 trillion** across 376 events. National Oceanic and Atmosph…

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## Turning these numbers into compelling visualizations

The data above supports several high-impact visualization formats. The carbon pool comparison (soils vs. atmosphere vs. vegetation) translates directly into a proportional area chart or animated carbon-flow diagram. The 133 Pg C soil carbon debt provides a powerful before/after narrative, PNAS especially when paired with sequestration potential under regenerative management. The Rodale 34-year time series (conventional vs. organic-legume vs. organic-manure soil carbon) is ideal for an animated line chart showing divergence over decades. Wiley Online Library

For economics, the juxtaposition of **78% higher profits with 29% lower production** creates an intuitive two-bar comparison that challenges conventional thinking. Farm bankruptcy trends (2019–2025) and disaster frequency acceleration (82-day vs. 18-day intervals between billion-dollar events) offer compelling time-series animations. The Central Valley water story — subsidence maps, almond acreage growth overlaid with water table decline, and the projected 87% delivery reduction — provides geographic and temporal visualization opportunities uniquely relevant to The Edison Institute's location.

The most powerful cross-pillar insight is the interconnection: rebuilding 1% soil organic matter simultaneously sequesters carbon, holds thousands of additional gallons of water, reduces input costs, and improves drought resilience. That single metric — **percent soil organic matter** — is the thread connecting all three pillars and could serve as the central interactive element across the entire visualization suite.